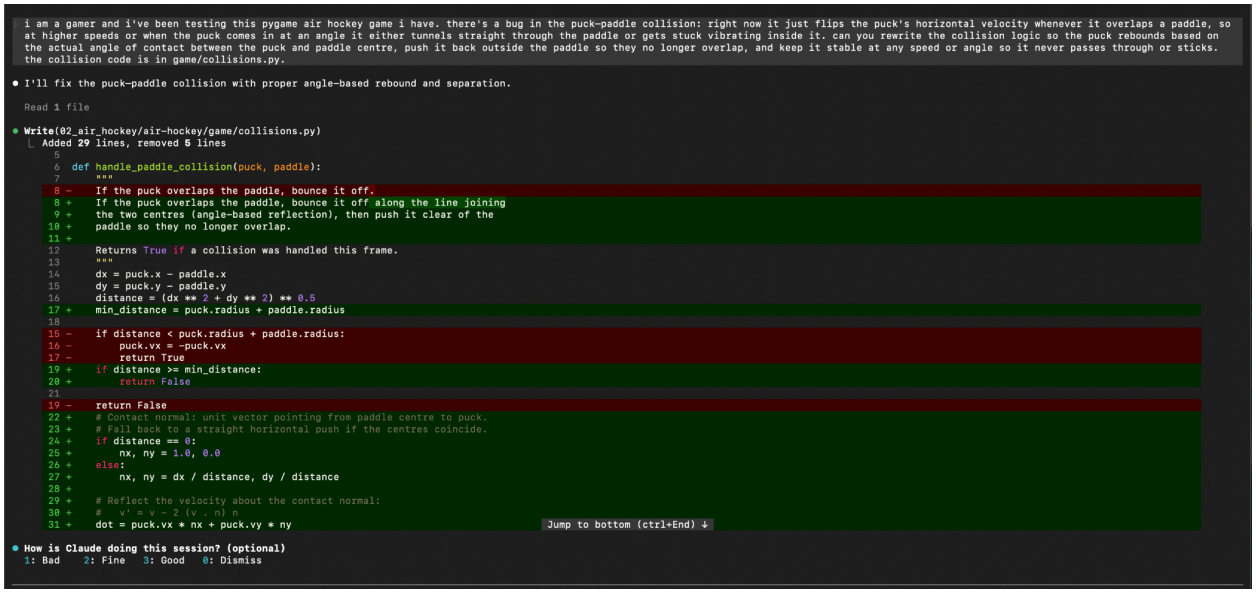
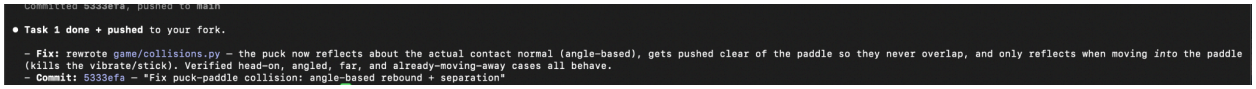
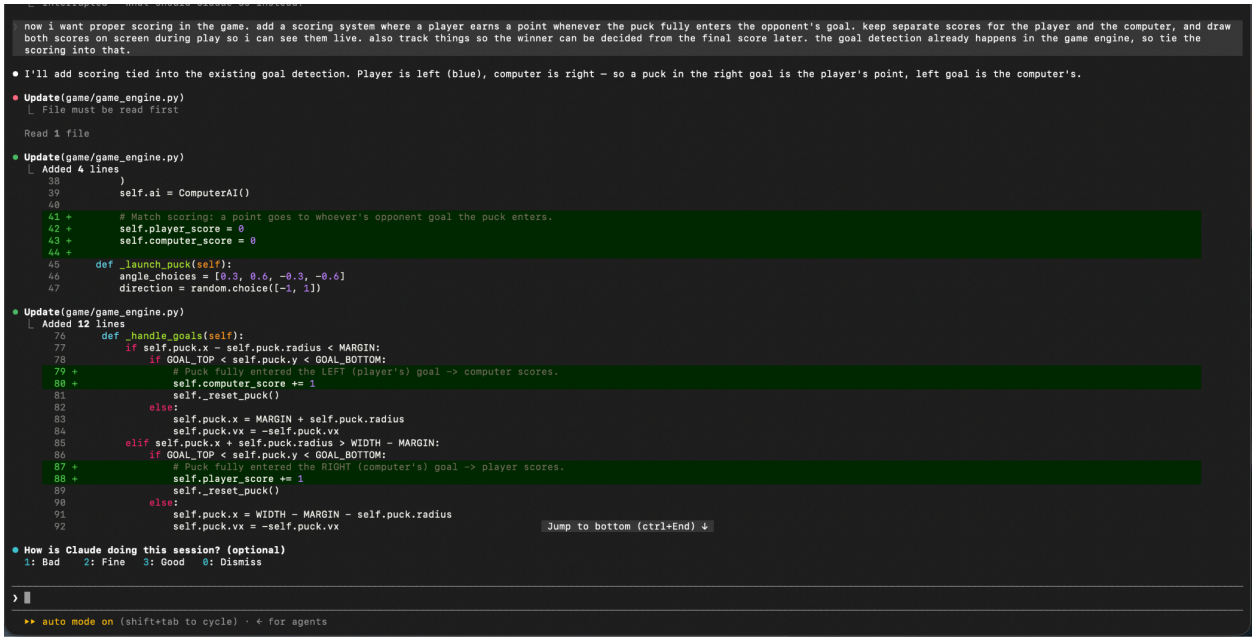
SE LAB 4
NAME - SAMYAK SANKLECHA
SRN - PES2UG24CS436
GITHUB - https://github.com/samyak-1021/PES2UG24CS436_02_air_hockey

Task 1 -





Task 2 -



● **Task 2 done + pushed.**

- **Added:** separate `player_score` / `computer_score`, a point awarded when the puck fully enters a goal (right = player, left = computer), both scores drawn live on-screen (blue left, red right), plus a `get_winner()` helper for later. Verified both goals score correctly and the winner/draw logic works.
- **Commit:** `b23969b` — "Add match scoring with live on-screen scores"

Task 3 -

> now i want each match to be time-limited. add a 30 second countdown timer that shows the remaining time on screen during gameplay. when the timer hits zero, stop the match (puck stops, no more scoring) and show the result on screen — whoever has the higher score wins, and if both scores are equal show "Draw". make sure pressing R restarts the timer along with everything else.now now

● I'll add the timer, the end-of-match freeze + result, and wire R to restart everything.

● **Update**(air-hockey/game/game_engine.py)
└ Added **4** lines

```
 7   a goal is incomplete. That's what Tasks 2–4 fix/add.
 8   """
 9
10 +import math
11   import random
12
13 +import pygame
14 +
15   from game.puck import Puck
16   from game.paddle import Paddle
17   from game.ai import ComputerAI
...
22   PUCK_RADIUS = 12
23   PADDLE_RADIUS = 28
24   INITIAL_PUCK_SPEED = 4.5
25 +MATCH_DURATION = 30  # seconds
26
27
28   class GameEngine:
```

● **Update**(air-hockey/game/game_engine.py)
└ Added **14** lines

```
46           self.player_score = 0
47           self.computer_score = 0
48
49 +         # 30-second match timer.
50 +         self.game_over = False
51 +         self.winner = None
52 +         self.match_start_ms = pygame.time.get_ticks()
53 +
54 +     def reset(self):
55 +         """Restart the whole match: scores, puck, paddles and timer."""
56 +         self.__init__()
57 +
58 +     def time_left(self):
59 +         """Seconds remaining in the match (never below 0)."""
```

Jump to bottom (ctrl+End) ↓

● **How is Claude doing this session? (optional)**
1: Bad 2: Fine 3: Good 0: Dismiss

Image in clipboard · ctrl+v to paste

Committed `fe66ab8`, pushed to main

● **Task 3 done + pushed.**

- **Added:** a 30-sec countdown shown centered at the top; at zero the puck freezes, scoring stops, and it locks in the winner (or "Draw") with a result banner + "Press R to restart" hint. Wired **R** to reset scores, puck, paddles *and* the timer. Verified all of it.
- **Commit:** `fe66ab8` — "Add 30-second match timer with end-of-match result and R restart"

Task 4 -

> right now after a goal the puck just freezes dead in the centre because its velocity gets set to zero. fix the reset so after every goal the puck goes back to the centre and immediately re-launches on its own with a fresh random direction and the normal starting speed, so the next point starts cleanly and the puck never sits motionless.

● Simple fix — reposition to center then re-launch with a fresh random direction:

● **Update**(air-hockey/game/game_engine.py)
└ Added **3** lines, removed **2** lines

```
133          return "Draw"
134
135      def _reset_puck(self):
136 +        # Send the puck back to the centre and immediately re-launch it with a
137 +        # fresh random direction at the normal speed, so it never sits still.
138          self.puck.x, self.puck.y = WIDTH / 2, HEIGHT / 2
137 -        self.puck.vx = 0
138 -        self.puck.vy = 0
139 +        self._launch_puck()
140
141      def draw(self, surface, font):
142          from game import renderer
```

Ran **1** shell command

● Works — puck returns to center and immediately relaunches moving, at the normal speed. Committing + pushing (last task).

Committed `c65f3b9`, pushed to main

● **Task 4 done + pushed. All 4 tasks complete!** 🎉

- **Fix:** puck now returns to center and immediately re-launches with a fresh random direction at normal speed — never sits dead.
- **Commit:** `c65f3b9` — "Reset and relaunch puck to centre after each goal"
- **Author:** you only ✅

Summary of what's on your fork now

Task	Commit	What
1 (bug)	`5333efa`	Fixed puck–paddle collision (angle rebound + separation)
2	`b23969b`	Match scoring + live on-screen scores
3	`fe66ab8`	30-sec timer + result banner + R restart
4	`c65f3b9`	Puck resets & relaunches after each goal

Jump to bottom (ctrl+End) ↓

● **How is Claude doing this session? (optional)**
1: Bad 2: Fine 3: Good 0: Dismiss